



CANDIDATE
NAME

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CENTRE
NUMBER

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NUMBER

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0620/42

October/November 2020

1 hour 15 minutes

You must answer on the question paper.

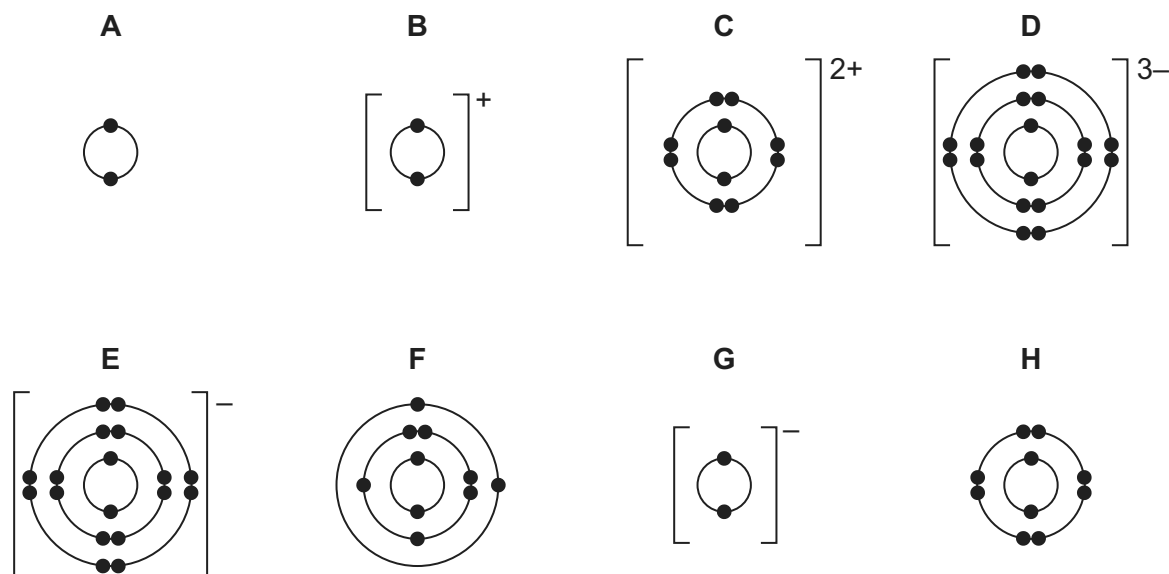
No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [].
- The Periodic Table is printed in the question paper.

This document has **12** pages. Blank pages are indicated.

1 The electronic structures of some atoms and ions are shown.



(a) Write the letters, **A**, **B**, **C**, **D**, **E**, **F**, **G** or **H**, of the electronic structures which show:

(i) atoms of two different noble gases and [2]

(ii) an ion of a Group I element [1]

(iii) an ion of a Group V element [1]

(iv) a pair of ions that could form a compound with the formula XY_2 and [1]

(b) State which electronic structure, **A**, **B**, **C**, **D**, **E**, **F**, **G** or **H**, is incorrect.

Explain why.

incorrect electronic structure

explanation

[2]

(c) State how many protons are found in the nucleus of ion **C**. [1]

(d) Use the Periodic Table to deduce:

(i) the chemical symbol for ion **G** [1]

(ii) the element which forms an ion with a 3+ charge and the same electronic structure as **H**.

..... [1]

[Total: 10]

2 Soluble salts can be made by adding a metal carbonate to a dilute acid.

(a) Give the formula of the dilute acid which reacts with a metal carbonate to form a nitrate salt.

..... [1]

(b) A student wanted to make hydrated iron(II) sulfate crystals, $\text{FeSO}_4 \cdot x\text{H}_2\text{O}$, by adding excess iron(II) carbonate to dilute sulfuric acid. The student followed the procedure shown.

step 1 Add dilute sulfuric acid to a beaker.

step 2 Add small amounts of iron(II) carbonate to the dilute sulfuric acid in the beaker until the iron(II) carbonate is in excess.

step 3 Filter the mixture formed in **step 2**.

step 4 Heat the filtrate until it is a saturated solution. Allow to cool.

step 5 Once cold, pour away the remaining solution. Dry the crystals between filter papers.

(i) Why must the iron(II) carbonate be added in excess in **step 2**?

..... [1]

(ii) State **two** observations in **step 2** that would show that iron(II) carbonate was in excess.

1

2

[2]

(iii) Describe what should be done during **step 3** to ensure there is a maximum yield of crystals.

..... [1]

(iv) A saturated solution is formed in **step 4**.

Describe what a saturated solution is.

.....

..... [2]

(v) Name a different compound that could be used instead of iron(II) carbonate to produce hydrated iron(II) sulfate crystals from dilute sulfuric acid.

..... [1]

- (c) On analysing the crystals, the student found that one mole of the hydrated iron(II) sulfate crystals, $\text{FeSO}_4 \cdot x\text{H}_2\text{O}$, had a mass of 278 g.

Determine the value of x using the following steps:

- calculate the mass of one mole of FeSO_4

mass = g

- calculate the mass of H_2O present in one mole of $\text{FeSO}_4 \cdot x\text{H}_2\text{O}$

mass of H_2O = g

- determine the value of x .

x =
[3]

- (d) Insoluble salts can be made by mixing solutions of two soluble salts.

A student followed the procedure shown to make silver bromide, an insoluble salt.

step 1 Add aqueous silver nitrate to a beaker. Then add aqueous potassium bromide and stir.

step 2 Filter the mixture formed in **step 1**.

step 3 Dry the residue.

- (i) State the term used to describe this method of making salts.

..... [1]

- (ii) Give the observation the student would make during **step 1**.

..... [1]

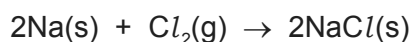
- (iii) Write the ionic equation for the reaction between aqueous silver nitrate and aqueous potassium bromide.

Include state symbols.

..... [3]

- (e) Sodium chloride is an ionic salt. It can be made by reacting sodium with chlorine gas.

The equation for this reaction is shown.



Calculate the volume of chlorine gas, in cm^3 , that reacts to form 2.34 g of NaCl.

The reaction takes place at room temperature and pressure.

volume of chlorine gas = cm^3 [3]

- (f) Sodium chloride does not conduct electricity when solid, but does conduct electricity when molten.

- (i) Explain why, in terms of structure and bonding.

.....

 [3]

- (ii) Name the product formed at the positive electrode when electricity is passed through molten sodium chloride.

..... [1]

- (iii) State the type of change that occurs at the positive electrode in (ii).

Explain your answer in terms of electron transfer.

type of change
 explanation [2]

- (iv) Describe what else can be done to sodium chloride to allow it to conduct electricity.

..... [1]

[Total: 26]

- 3 Group I metals are very reactive. Transition elements are also metals but are less reactive than Group I metals.

(a) State **two** physical properties of Group I metals which are similar to those of transition metals.

1

2 [2]

(b) Describe **two** ways in which the physical properties of Group I metals are different from those of transition metals.

1

.....

2

..... [2]

(c) When Group I metals are added to water they fizz and an alkaline solution forms.

(i) Name the gas given off.

..... [1]

(ii) Identify the ion present in the solution which makes the solution alkaline.

..... [1]

(iii) Write the chemical equation for the reaction between sodium and water.

..... [2]

(d) When the transition element iron is added to water the iron rusts.

When an iron object is coated with a layer of zinc, rusting is prevented.

(i) Name this process of coating iron objects with a layer of zinc.

..... [1]

(ii) Explain how completely coating an iron object with a layer of zinc prevents rusting.

..... [1]

(iii) Rusting of iron ships can be prevented by attaching zinc blocks to the hull of the ship.

Explain how this prevents rusting.

.....

..... [2]

[Total: 12]

4 Alkenes and alkanes are homologous series of compounds containing carbon and hydrogen atoms.

(a) State the name of the type of compound made from carbon and hydrogen atoms only.

..... [1]

(b) Alkenes take part in addition reactions.

(i) Describe what is meant by the term *addition reaction*.

..... [1]

(ii) Draw the structure of the product made in the addition reaction between propene and bromine. Show all of the atoms and all of the bonds.

[2]

(iii) Describe the colour change seen when propene is added to aqueous bromine.

from to [2]

(iv) Draw the structures of molecules of **two** different alkenes which both undergo an addition reaction with steam to form butan-2-ol. Show all of the atoms and all of the bonds.

[2]

(c) Propane undergoes a substitution reaction with chlorine.

Write the chemical equation for the reaction between one molecule of propane and one molecule of chlorine.

..... [2]

[Total: 10]

5 This question is about alcohols, carboxylic acids and esters.

(a) Ethanol will react with hot aqueous potassium manganate(VII) to form ethanoic acid.

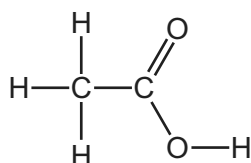
(i) State the other condition needed for this reaction to take place.

..... [1]

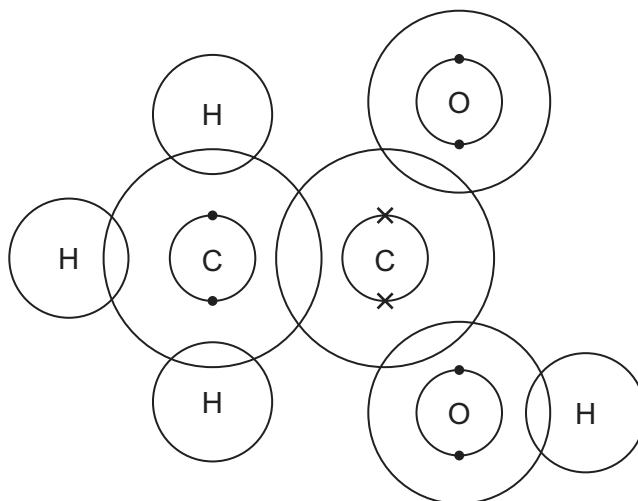
(ii) State the type of chemical change that happens to the ethanol during this reaction.

..... [1]

(iii) The structure of ethanoic acid is shown.



Complete the dot-and-cross diagram to show the electron arrangement in a molecule of ethanoic acid.



[3]

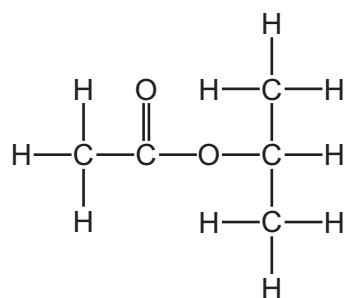
- (b) Ethanoic acid is a weak acid and hydrochloric acid is a strong acid.

Complete the table to show the similarities and differences in the properties of samples of these two acids of equal concentration.

	dilute ethanoic acid	dilute hydrochloric acid
extent of dissociation		
colour after adding universal indicator solution		
observation when magnesium ribbon is added		

[6]

- (c) Ethanoic acid will react with an alcohol to form the ester shown.



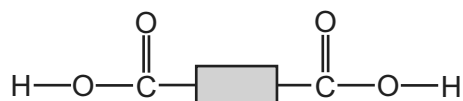
- (i) Name the **other** product formed when ethanoic acid reacts with an alcohol to make this ester.
 [1]
- (ii) Give **one** condition needed when ethanoic acid reacts with the alcohol to make this ester.
 [1]
- (iii) Draw the structure of the alcohol which was added to ethanoic acid to make this ester. Show all of the atoms and all of the bonds.

[2]

- (d) Polyesters can be manufactured from carboxylic acids and alcohols.

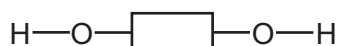
Hexanedioic acid has the structure: $\text{HOOC}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{COOH}$.

This structure can be simplified as shown.



Ethanedioic acid has the structure: $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$.

This structure can be simplified as shown.



The functional groups are found at the end of each molecule.

- (i) State what is meant by the term *functional group*.

..... [1]

- (ii) Determine the empirical formula of hexanedioic acid.

..... [1]

- (iii) Calculate the percentage by mass of oxygen present in ethanedioic acid.

Give your answer to the nearest whole number.

..... % [2]

- (iv) Complete the diagram to show a section of polyester manufactured from hexanedioic acid and ethanedioic acid. Include all of the atoms and all of the bonds in the linkages.



[2]

- (v) State the name of a polyester.

..... [1]

[Total: 22]

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The Periodic Table of Elements

Group															
I	II							III	IV	V	VI	VII	VIII		
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										2 He helium 4			
11 Na sodium 23	12 Mg magnesium 24	1 H hydrogen 1	5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20							
19 K potassium 39	20 Ca calcium 40	13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40								
37 Rb rubidium 85	38 Sr strontium 88	30 Zn zinc 65	29 Cu copper 64	28 Ni nickel 59	27 Co cobalt 59	26 Fe iron 56	25 Mn manganese 55	24 Cr chromium 52	23 V vanadium 51	22 Ti titanium 48	21 Sc scandium 45	54 Xe xenon 131	86 Rn radon —		
55 Cs caesium 133	56 Ba barium 137	80 Hg mercury 201	79 Au gold 197	78 Pt platinum 195	77 Ir iridium 192	76 Os osmium 190	75 Re rhenium 186	74 W tungsten 184	73 Ta tantalum 181	72 Hf hafnium 178	71 La lanthanum 139	85 At astatine —	—		
87 Fr francium —	88 Ra radium —	112 Cn copernicium —	111 Rg roentgenium —	110 Ds darmstadtium —	109 Mt meitnerium —	108 Hs hassium —	107 Bh bohrium —	106 Sg seaborgium —	105 Db dubnium —	104 Rf rutherfordium —	89–103 actinoids —	116 Lv livermorium —	—		

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).